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SEAT No. :

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[6181]-172

B.E. (Electronics /E& TC)

REMOTE SENSING

(2019 Pattern) (Semester - VIII) (Elective - VI) (404192C) (Theory)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q5 or Q6, Q.7 or Q.8.
- 2) Figures to the right indicate full marks.

- Q1)** a) Explain the Geometric corrections and atmospheric corrections and how they are removed during the pre-processing stage of satellite images.[8]
- b) Explain various value added products offered by the space agencies to the end user for satellite data processing. [8]

OR

- Q2)** a) Write short notes on : [8]
- i) True Color Composite
- ii) False Color Composite
- b) Describe the significance of NDVI. Calculate NDVI of a plant with NIR = 0.50 and RED = 0.08 and determine whether the plant is healthy or stressed. [8]
- Q3)** a) Differentiate between active and passive microwave remote sensing. [8]
- b) What is SAR and SLAR. Brief about the characteristics and interpretation of SAR data. [10]

OR

- Q4)** a) Explain Radar equation and determine ground based resolution and azimuth resolution of SLAR with suitable illustrations. [8]
- b) Explain LIDAR concept and its key characteristics with applications.[10]

P.T.O.

- Q5) a)** Name India's own navigation system and its key features and application areas with suitable diagram of satellite positioning. **[10]**
- b)** Explain the principles of satellite positioning and GNSS orbits. **[8]**

OR

- Q6) a)** What is an augmentation system? Differentiate between Space based and Ground based augmentation system. **[10]**
- b)** Write short notes on LBS with suitable real life examples. **[8]**
- Q7) a)** Brief the application of remote sensing in Landuse and Land cover with use cases and methodology. **[8]**
- b)** Brief the application of remote sensing in water resource management with use cases and methodology. **[10]**

OR

- Q8) a)** Brief the application of remote sensing in precision agriculture with use cases and methodology. **[10]**
- b)** Brief the application of remote sensing in Urban planning and management with use cases and methodology. **[8]**

